Natural Language Processing of Clinical Notes for Improved Early Prediction of Septic Shock in the ICU

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***Abstract*— Sepsis and septic shock are major concerns in public health as the leading contributors to hospital mortality and cost of treatment in the United States. Early treatment is instrumental for improving patient outcome; to this end, algorithmic methods for early prediction of septic shock have been developed using electronic health record data, with the goal of decreasing treatment delay. We extend a previously-developed method, using a gradient boosting algorithm (XGBoost) to compute a time-evolving risk of impending transition into septic shock, by combining physiological data from the electronic health record with features obtained from natural language processing of clinical note data. We compare two different methods for generating natural language processing features, with the best method obtaining improved performance of 0.92 AUC, 84% sensitivity, 82% specificity, 49% positive predictive value, and a median early warning time of 7.0 hours. This degree of early warning is sufficient to enable intervention many hours in advance of septic shock onset, with the improved prediction performance of this method resulting in fewer false alarms and thus more actionable predictions.**

I. INTRODUCTION

Sepsis is a syndrome, or group of symptoms, associated with infection, rather than a single disease [1]. Understanding of the underlying pathophysiology of sepsis has changed over time [2]; the first set of consensus definitions (Sepsis-1) were laid out in 1991 [3], defining sepsis as systemic inflammatory response syndrome (SIRS) caused by infection. A second consensus conference (Sepsis-2) in 2001 [4] expanded the list of signs and symptoms associated with sepsis, but continued to use SIRS criteria for diagnosis of sepsis. Most recently, in 2016, the Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3) [1] defined Sepsis as life-threatening organ dysfunction caused by dysregulated host response to infection, and septic shock as a subset of sepsis in which profound circulatory, cellular, and metabolic abnormalities result in substantially greater mortality. These changing definitions illustrate that sepsis is a disease which is not yet fully understood, and is at times vaguely defined.

Nonetheless, sepsis and septic shock pose major concerns in public health as the leading causes of hospital mortality [5] and the costliest medical conditions across all payers in the United States [6]. Mortality in septic shock is estimated to be as high as 45% [7], [8]. Though treatment recommendations for sepsis are subject to the same continual revisions as the diagnostic criteria [9], [10], [11], it is clear that timely treatment is instrumental in improving patient outcome: Kumar et al. first found in 2006 that septic shock patients treated within the first hour of diagnosis had a survival rate of 80%, but each hour of delayed treatment increased mortality by 8% [12]. Though more recent studies have corroborated this finding of increased mortality with delayed treatment, delays in treatment remain common in current practice [13], [14], [15]. To this end, we previously developed a method for early prediction of septic shock using physiological data from the electronic health record (EHR) [16].

However, a great deal of EHR data is contained within unstructured clinical notes or narrative entries. These notes contain clinician insight into the course of treatment and status of the patient, information which is not available in the values of physiological variables in the EHR. A number of studies have sought to employ natural language processing (NLP) techniques to extract predictive features from these clinical notes for problems such as prediction of mortality in ICU patients, or topic classification [17], [18], [19]. However, the majority of these studies consider these NLP features in isolation, rather than in conjunction with physiological data. Recently, Marafino et al. combined NLP features with EHR data to predict mortality in the ICU [20]. In this study, they demonstrate that a combination of NLP and physiological features achieves greater classification performance than NLP or physiological features alone.

We present a first approach to employing NLP features for early prediction of septic shock: we extend our previously developed method for early prediction of septic shock by combining features from NLP with physiological data from the EHR, and achieve improved performance using this combination of NLP and EHR features over our previous method with EHR data alone. These features are calculated on a time-evolving basis, using the most recently observed values for both notes and structured physiological data, making this method suitable for prospective study and implementation in real time.

II. METHODS

A. Data extraction and processing

The MIMIC-III clinical database (version 1.4) [21] contains EHR data for 38,645 adult patients collected between 2001-2012. 28 physiological variables for these patients were extracted from the MIMIC-III PostgreSQL database according to the procedure previously described [16]. 70% of patients were randomly sampled into the training set, whereas the remaining 30% were reserved for testing. Over 2 million clinical notes are contained within this database. These were extracted from the noteevents table. All notes were converted to lowercase and stripped of non-alphanumeric characters.

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B. Labeling clinical state

We apply the Sepsis-3 definitions [1] to determine clinical state (non-sepsis, sepsis, or septic shock) for each patient at each time point where there is a new observation of physiological data. Missing values are imputed by carrying forward the last observation, where available; if no previous observation is available, then it is assumed that the patient does not meet disease criteria for that variable. Suspected infection is determined according to Seymour et al. [22] using concomitant orders for antibiotics and blood cultures. A patient has sepsis if they have suspected infection and a SOFA score of 2 or higher. Patients are in septic shock when they have sepsis, have received adequate fluid resuscitation, require vasopressors to maintain a mean arterial blood pressure of at least 65 mmHg, and have a serum lactate >2mmol/L. Adequate fluid resuscitation follows the treatment guidelines of the 2012 Surviving Sepsis Campaign [10]: a patient is considered adequately fluid resuscitated if they have received at least 30 mL/kg of fluids, or if the treatment targets of urine output >0.5 mL/kg/hr or CVP of 8-12 mmHg have been met. Time of septic shock onset is determined as the earliest time when all the criteria for septic shock are met.

C. Generation of NLP features

Text processing for the generation of NLP features was conducted using the *text2vec* R package [23]. First, we generate a vocabulary by computing a list of all unique words (alphanumeric strings delimited by whitespace). This vocabulary is pruned by eliminating terms which occur less than once per 1000 notes (occurrence is too rare), and terms which occur in more than half of the notes (occurrence is too frequent to have predictive value; e.g. articles and prepositions such as the, a, an, these, of, etc.). This yields a vocabulary of 8907 unique terms.

We compare two different methods for generating NLP features. We first compute the document-term matrix by counting the number of occurrences of each term in the pruned vocabulary in each note, resulting in a matrix where the i, j-th element is the number of occurrences of term j in note i. Further feature selection was done by training a gradient boosting model (XGBoost) [24] for classification of notes as being from a patient who never entered septic shock, or from a patient who eventually entered septic shock from a timestamp preceding septic shock onset. A total of 832 terms were selected as predictive of impending transition from sepsis to septic shock, and used in addition to physiological data for early prediction of septic shock.

The second method uses recurrent neural networks and word embeddings for feature extraction. Global Vectors for Word Representation (GloVe) [25] is used to learn a 300-dimensional vector embedding of each of the 8907 words in the pruned vocabulary. Each note is represented as a matrix, where the i-th row contains the embedding of the i-th word in the note; these matrices were computed using the first 1000 words of each note, and were padded with 0s for notes shorter than 1000 words. A recurrent neural network was then trained to classify each sequence as being from a patient who never entered septic shock, or from a patient who eventually entered septic shock from a timestamp preceding septic shock onset. Figure 1 illustrates the structure of the recurrent neural network. Training data for the neural network was rebalanced by upsampling data from septic shock patients to match the number of occurrences of data from non-shock patients. The values of the last hidden layer are used as features, resulting in a 128-dimensional vector representation of each note. These features computed from the recurrent neural network were used in addition to physiological data for early prediction of septic shock.

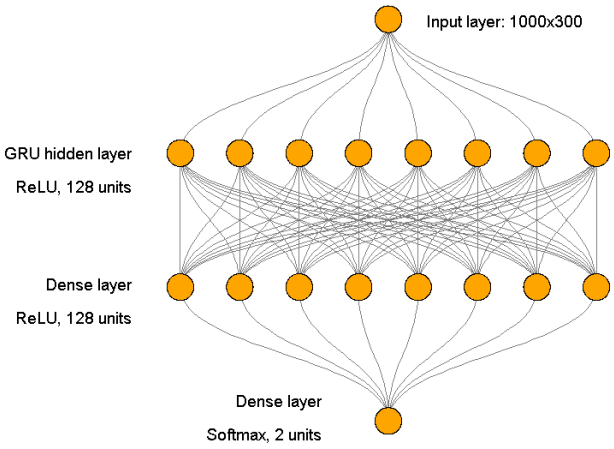

Fig. 1. Structure of neural network used for NLP feature generation, consisting of two intermediate layers with rectified linear unit (ReLU) activation. Notes are represented by a 1000×300 matrix, where row i is the word vector for the ith word in the note. After training, the output of the last hidden layer (128-unit dense) is used as a vector representation of each note.

D. Risk Modeling

The basis of our early prediction method is the assumption that there exists a pre-shock state [16]. Entry into this state is indicative of impending transition into septic shock. We train a gradient boosting model (XGBoost) [24] for regression between data from patients who never enter septic shock, and data from the pre-shock state. The output of this model is a risk score, quantifying the relative likelihood that a patient will ultimately develop septic shock. Training data is generated by sampling a fixed number of data points from each patient in the training set. To reduce computational complexity, we select 10 data points per patient for training. For non-shock patients, this data is selected by uniformly sampling time points from the span of observations which are available, then evaluating the value of each feature at each timestamp. For shock patients, this data is sampled from the time interval spanning 2 hours prior to septic shock onset to 1 hour prior to septic shock onset.

When only physiological data is used for early prediction of septic shock, these features were calculated by using the most recent observation of each physiological variable wherever available. If no observation is available, mean value imputation is used. When both physiological and

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NLP features are used, the physiological feature vector is concatenated with the computed NLP features of the most recent clinical note. If the document-term matrix is used to compute NLP features, the physiological feature vector is concatenated with the vector containing the occurrences of the 832 selected terms. If word embeddings and recurrent neural networks are used to compute NLP features, the physiological feature vector is concatenated with the 128-dimensional vector computed by the neural network. If no clinical note is available, mean value imputation is used for the document-term vector, and a matrix of all 0s is used as input to the neural network. Because timestamps for clinical notes only indicate the date and do not specify the time of entry, we imposed the condition that the date of the note must precede the date of the current timestamp, to ensure that data from notes entered subsequent to the current timestamp are not used.

E. Prediction

For each patient, we apply the gradient boosting model to each timepoint at which there is a new observation in the EHR to compute a time-evolving risk score. The first time at which this risk score exceeds a fixed threshold is when we predict that the patient has entered the pre-shock state, and thus is likely to develop septic shock. A patient whose risk score exceeds the threshold prior to septic shock onset and ultimately develops septic shock is a true positive. A patient whose risk score never exceeds the threshold and does not develop septic shock is a true negative. A patient whose risk score does not exceed the threshold prior to septic shock onset and develops septic shock is a false negative. A patient whose risk score exceeds the threshold but does not develop septic shock is a false positive. Early warning time is defined as the difference between the time of threshold crossing and the time of septic shock onset. All model selection and training is conducted using the training set. Detection threshold was learned from the training set by selecting the point closest to the top left-hand corner of the ROC curve.

We apply our method for early prediction of septic shock to the 15,930 adult patients in MIMIC-III with suspected infection and at least one entry of physiological EHR data. Confidence intervals were computed using bootstrap. Fig. 2 illustrates examples of the risk score trajectory for a patient who develops septic shock, and a patient who never develops septic shock.

III. RESULTS

With the addition of NLP features, we obtain significantly improved AUC, sensitivity, and early warning time for early prediction of septic shock compared to using only physiological variables from the EHR. Table I lists performance for early prediction of septic shock on the test set using only physiological variables, as well as physiological variables in conjunction with NLP features. Performance that is significantly greater than that achieved using only physiological data at the 0.01 significance level is indicated with an asterisk (*). Performance that is significantly greater (using the one-sided Wilcoxon rank-sum test) than that achieved using only physiological data at the 0.05 significance level is indicated with a dagger (†). Fig. 3 shows ROC curves for early prediction with and without the addition of NLP features. Though the highest AUC is obtained using GloVe embeddings and recurrent neural networks, and is significantly greater than the AUC obtained using the document-term matrix (DTM) for NLP feature generation (p=2.61e-4), we rely upon results using the document-term matrix to interpret the contribution of NLP features to improved performance in early prediction, as these features are more readily interpretable than those produced by the deep learning-based method.

Relative feature importance for XGBoost can be calculated

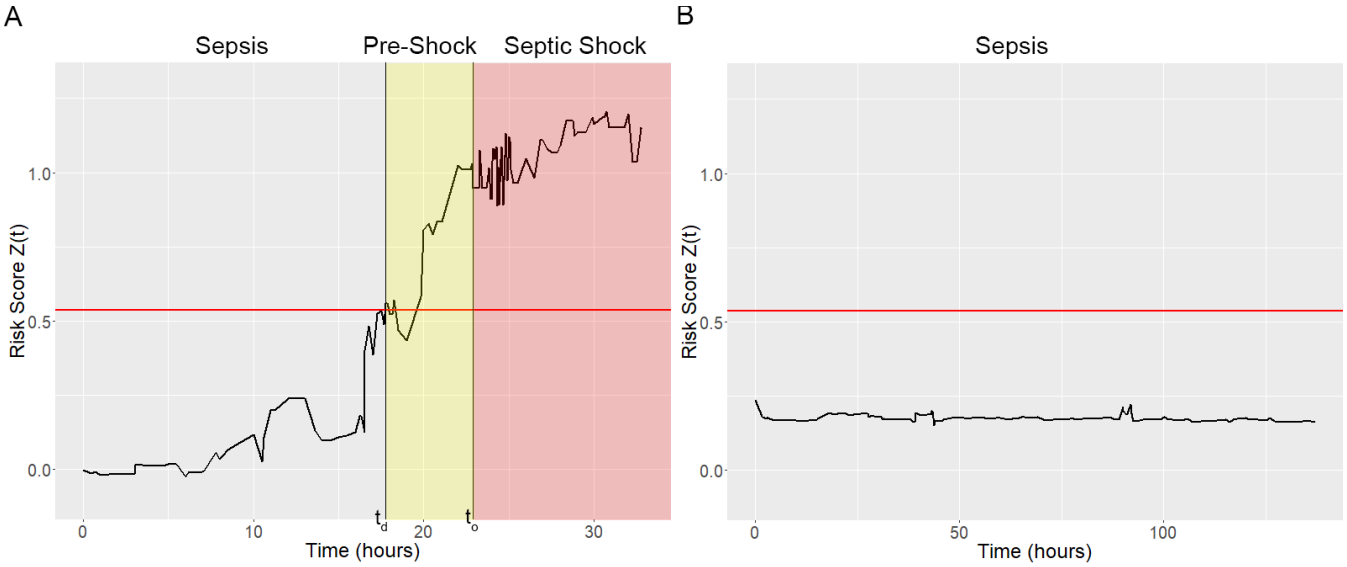

Fig. 2. Illustration of early prediction method for a patient who (A) enters septic shock and a patient who (B) never enters septic shock. We predict that the patient has entered pre-shock when their risk score exceeds the threshold (red line).

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TABLE I
PERFORMANCE METRICS

Performance Metric	Physiologic Only	Physiologic + DTM	Physiologic + GloVe/GRU
AUC	0.85	0.91*	0.92*
Sensitivity	0.76	0.86*	0.84*
Specificity	0.79	0.79	0.82*
PPV	0.43	0.46†	0.49*
Median EWT (hrs)	6.0	7.3*	7.0*

TABLE II
TOP 25 FEATURES IN XGBOOST MODEL

Feature	Importance
Cardiovascular SOFA	0.3135
Lactate	0.2398
Heart Rate	0.0819
Systolic BP	0.0721
GCS	0.0439
Arterial pH	0.0391
Creatinine	0.0256
WBC	0.0209
Respiratory Rate	0.0160
FiO2	0.0140
Bilirubin	0.0089
PaO2	0.0070
Potassium	0.0062
Temperature	0.0037
CVP	0.0031
PaCO2	0.0030
Hematocrit	0.0028
Hemoglobin	0.0022
”times”	0.0015
”patient”	0.0015
Platelets	0.0014
”liver”	0.0013
BUN	0.0013
”given”	0.0012
Urine Output	0.0012

as the average improvement in training performance when using a feature for splitting a decision tree. Table II lists the top 25 physiologic and NLP features used to generate the risk score for early prediction of septic shock. NLP features are given in quotation marks. We see that of the 28 physiological features, 18 take precedent over any NLP feature, indicating that the contribution of individual terms is low relative to physiological variables. Two physiological features, Cardiovascular SOFA score and Lactate have a substantially higher contribution to performance than the remainder of features. Table III contains a list of the 25 most predictive terms for classifying pre-shock notes vs non-shock notes.

IV. DISCUSSION

A. NLP Features

If we examine Table III, we note that the XGBoost model, used for feature selection, captures terms related to sepsis, septic shock and treatments. However, the most predictive terms from the document-term matrix are not among the most predictive features in our combined model using physiological data and the document-term matrix. It is likely that the information captured by these terms is at least partially reflected in the physiological data, and thus, some of these data are redundant. Nonetheless, the addition of NLP features substantially increases the sensitivity of our early prediction of septic shock, and also improves median early warning time. Clinical notes contain clinician insight into patient state, as well as the course of treatment; incorporating this information into predictive models can yield improved performance.

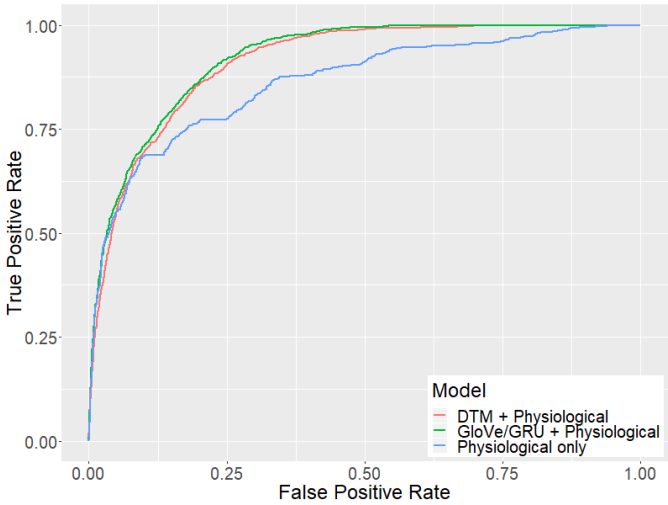

Fig. 3. ROC plots for early prediction with NLP features + physiological data and physiological data alone

The procedure for generating NLP features in this study is a rudimentary process, capturing only the frequency of occurrence of a certain set of words. No information pertaining to syntax or sentence structure is captured. The current state of the art in NLP revolves largely around deep learning-based embeddings [26] [27] [28]. These models are able to capture sentence structure, as well as form geometric representations of semantic meaning within text (e.g. words or sentences with similar meaning can be embedded such that the Euclidean distance between their vector representations is smaller than unrelated words) [26]. These models for deriving features from text may generate more meaningful representations of the contents of clinical notes, and thus enable greater predictive performance to be achieved. We obtain increased AUC using our GloVe/GRU-based method for NLP feature generation compared to using the document-term matrix, indicating that this method is indeed able to extract information from structure and syntax of clinical notes not present in the document-term matrix; moreover, the NLP feature vector is reduced in dimensionality from 832 terms to 128 features. Deep learning-based methods require careful tuning of hyperparameters to achieve optimal performance, and it is possible that a different network structure, or different values for hyperparameters may result in further improved performance. However, this increase in performance, while statistically significant, may not be large

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TABLE III
NLP TERMS MOST PREDICTIVE OF PRE-SHOCK STATE

Term	Importance
”tube”	0.0603
”crrt”	0.0514
”ards”	0.0385
”vasopressin”	0.0348
”portable”	0.0343
”failure”	0.0333
”shock”	0.0304
”sepsis”	0.0281
”dl”	0.0247
”levophed”	0.0245
”pressors”	0.0217
”norepinephrine”	0.0194
”action”	0.0194
”reason”	0.0169
”cvvh”	0.0159
”swan”	0.0123
”catheter”	0.0118
”admitting”	0.0103
”hepatorenal”	0.0098
”epinephrine”	0.0079
”sounds”	0.0073
”pancreatitis”	0.0071
”amylase”	0.0067
”arrest”	0.0065
”assessment”	0.0063

enough to outweigh the loss of interpretability associated with deep learning-based methods for feature extraction.

B. Prediction Method

The goal of any computational method for the early prediction of septic shock is to be able to alert clinicians when patients are at high risk of developing septic shock prior to onset, thus enabling timely intervention leading to improved patient outcome. The addition of NLP features to physiological data in the EHR yields improved performance in early prediction, as well as improved early warning time, compared to predictions made using physiological data alone. A method with greater early warning time allows for earlier therapeutic intervention, which has been shown to result in improved patient outcome [12], [13], [14], [15]. Improved prediction performance leads to predictions which are more actionable, with higher positive predictive value indicating a lower fraction of false alarms . This metric is of particular importance in practical application, as clinicians in the ICU are inundated with alarms on a constant basis.

C. Limitations

While the Sepsis-3 criteria reflect the latest understanding of the pathophysiology of sepsis, and recent studies have shown that Sepsis-3 labels more accurately reflect patient outcomes than Sepsis-2 [29], the adoption of these diagnostic criteria remains a point of debate. As the clinical states in this study are the result of applying these criteria, limitations of these labels become limitations of this study. Measurements of many different variables are required in order to calculate SOFA score; outside of the ICU, there may not be sufficient data to accurately evaluate the Sepsis-3 criteria. Another criticism of Sepsis-3 is that its criteria for diagnosing septic shock are treatment-based, and thus septic shock determined using the Sepsis-3 criteria may reflect a treated state. Nonetheless, the ability to predict septic shock many hours in advance of intervention is highly beneficial, as doing so enables more timely intervention.

Another limitation arises from the fact that the MIMIC-III database [21] is a database comprised of data collected from a single hospital. We have previously shown that our pre-shock state-based paradigm for training models for early prediction of septic shock is able to generalize to a multi-center database with moderate performance [16]. However, as no external validation set containing clinical notes is available, it is not certain how well the methods presented here will generalize to new clinical notes; variations in the practice of writing clinical notes or in terminology across hospitals may or may not result in degraded performance.

V. Conclusions

We demonstrate the applicability of two different methods for computing NLP features on clinical notes from the electronic health record, and apply these features to achieve improved performance in early prediction of septic shock. Best performance of 0.92 AUC, 84% sensitivity, 82% specificity, 49% positive predictive value, and a 7.0 hour median EWT is achieved using NLP features generated using GloVe word embeddings, indicating that information about syntax and sentence structure in clinical notes may yield additional predictive value over the number of occurrences of words. This degree of performance yields a window of time in which clinicians may be able to achieve earlier intervention on sepsis patients at risk of impending transition to septic shock, and thus improve patient outcome. Similar methods combining NLP features computed from clinical note data with physiological data in the EHR may result in improved performance on a variety of prediction tasks compared to using physiological data alone.

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